

$$x^2 - y = 0$$

$$V_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad u_1 = \begin{bmatrix} 0 \\ -1/2 \end{bmatrix} \quad f_1 = 0$$

$$x^2 + 2x + 1 + y = 0$$

$$V_2 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad u_2 = \begin{bmatrix} 1 \\ 1/2 \end{bmatrix} \quad f_2 = 1$$

$$\begin{vmatrix} 1+\mu & 0 & \mu \\ 0 & 0 & \frac{\mu-1}{2} \\ \mu & \frac{\mu-1}{2} & \mu \end{vmatrix} = 0$$

Solve,

$$-\left(\frac{\mu-1}{2}\right) \begin{vmatrix} 1+\mu & 0 \\ \mu & \frac{\mu-1}{2} \end{vmatrix} = 0$$

$$\Rightarrow -\left(\frac{\mu-1}{2}\right)(1+\mu)\left(\frac{\mu-1}{2}\right) = 0$$

$$\Rightarrow \mu = 1 \text{ or } -1$$

So, common chord

For $\mu = -1$,

$$C_1 + \mu C_2$$

$$[x \ y] \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + (-1) \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \Rightarrow (x^2 - y) - 1(x^2 + 2x + 1 + y) = 0$$

$$\Rightarrow -2x - 2y - 1 = 0$$

$$\Rightarrow x + y = -1/2$$

$$\Rightarrow x = h + km$$

$$\Rightarrow \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -1/2 \\ 0 \end{bmatrix} + k \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$k = - \begin{bmatrix} 1 & -1 \end{bmatrix} \left[\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \end{bmatrix} + \begin{bmatrix} 0 \\ -1/2 \end{bmatrix} \right] +$$

$$\sqrt{\begin{bmatrix} 1 & -1 \end{bmatrix} \left[\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \end{bmatrix} + \begin{bmatrix} 0 \\ -1/2 \end{bmatrix} \right] + \frac{1}{4} \begin{bmatrix} 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \end{bmatrix}}$$

$$g(h) = \begin{bmatrix} -1/2 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} -1/2 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 0 & -1/2 \end{bmatrix} \begin{bmatrix} -1/2 \\ 0 \end{bmatrix} + 0$$

$$g(h) = \frac{1}{4}$$

$$\Rightarrow R = 0 \pm \frac{\sqrt{0^2 - 1/4}}{1} \rightarrow \text{Imaginary} \Rightarrow \text{No sol}^n$$

For $M=1 \rightarrow x^2$ term $\rightarrow$ Not a chord